Two Giant Black Holes May Be Circling Toward a Crash

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Scientists think nearly every large **galaxy** has a **supermassive black hole** at its center. These objects can weigh millions or even billions of times as much as the Sun. Astronomers are still not sure how they grow so big. One idea is that black holes **merge** with each other when their galaxies collide.

Until now, astronomers had not clearly found a pair of **supermassive black holes orbiting** very close together. A team led by Silke Britzen of the Max Planck Institute for Radio Astronomy in Germany now reports strong evidence of such a pair at the center of a **galaxy** called Mrk 501.

The black hole in Mrk 501 was already known for shooting out a **jet** of particles at nearly the speed of light. The team studied radio observations collected on dozens of days over about 23 years. They spotted a second **jet**, pointing in a different direction, and tracked how it moved. That second **jet** strongly suggests a second black hole is there. The two are too close together to be seen separately.

"We searched for it for so long, and then it came as a complete surprise," Britzen said.

The second **jet** seemed to sweep around the larger black hole again and again, like part of a regular **orbit**. In June 2022, light from the system also appeared to form a ring. Scientists call this an **Einstein ring**, which happens when gravity bends light.

The team calculated that the two black holes circle each other about every 121 days. Depending on their masses, they could **merge** in as little as 100 years. That is a very short time for space, and it could help scientists learn how giant black holes come together. The study has been accepted by the journal Monthly Notices of the Royal Astronomical Society.

Word Treasure

supermassive black hole -- A supermassive black hole is an enormous black hole, millions or billions of times heavier than the Sun, found at the center of a galaxy.

galaxy -- A galaxy is a huge group of stars, gas and dust held together by gravity.

jet -- In space, a jet is a narrow beam of particles blasted out at high speed from near a black hole.

orbit -- An orbit is the curved path one object takes as it travels around another.

Einstein ring -- An Einstein ring is a ring of light formed when gravity bends light coming from an object behind another object.

merge -- To merge means to combine into one.

Background you need

A black hole is a place in space where gravity is so strong that not even light can escape.

Supermassive black holes sit at the centers of large galaxies, and some shoot out powerful jets of particles that radio telescopes can detect.

When two galaxies collide, their central black holes are expected to move closer, orbit each other and eventually merge.

Because the two black holes in Mrk 501 are too close to see separately, astronomers used the motion of their jets as evidence.

Break it down

LEAD: Astronomers report strong evidence of two supermassive black holes orbiting each other
BACKGROUND: black holes may grow by merging when galaxies collide
KEY DISCOVERY: a second jet in galaxy Mrk 501
- +- WHO: team led by Silke Britzen, Max Planck Institute for Radio Astronomy
- +- HOW: radio observations on dozens of days over about 23 years
- +- EXTRA CLUE: an apparent Einstein ring in June 2022

RESULT: the pair orbits about every 121 days and could merge in as little as 100 years
OPEN QUESTION: What are the black holes' exact masses, and how fast will they merge?

Why it matters

Finding a close pair of giant black holes helps scientists test the idea that black holes grow by merging. A merger within about 100 years would be a rare chance to study how these giants come together.

Test what you remember

1. **Where are supermassive black holes thought to be found?**
   - (A) Only in small galaxies
   - (B) Inside the Sun
   - (C) Only near Earth
   - (D) At the centers of nearly all large galaxies
2. **What clue suggested a second black hole in Mrk 501?**
   - (A) A new planet
   - (B) A change in the Sun's color
   - (C) A bright comet
   - (D) A second jet that could be tracked
3. **About how many years of radio observations did the team study?**
   - (A) 100
   - (B) 121
   - (C) 2
   - (D) About 23
4. **What is an Einstein ring?**
   - (A) A circle of stars in a galaxy
   - (B) A ring of light formed when gravity bends light
   - (C) A ring of gas around a planet
   - (D) A kind of telescope
5. **How often do the two black holes seem to circle each other?**
   - (A) About every 121 days
   - (B) Every 23 years
   - (C) Every day
   - (D) Every 100 years
6. **Who led the research team?**
   - (A) Silke Britzen
   - (B) Pulin Li
   - (C) Peter Jordan
   - (D) Mark Procter

Answer key (try the quiz first!): 1: D 2: D 3: D 4: B 5: A 6: A

Questions to think about

MY THOUGHTS (at least 20 words):